AI ACADEMY · LEVEL 7 · GRADE 12 · WEEK 29 · STUDENT LESSON

Figures that preserve evidence

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Draw the comparison with honest scales and visible denominators

Preserve units, case counts and uncertainty limits without making a figure stand in for missing evidence.

Keep one habit: Question and contract → complete main ledger → separate picture → boundary probes → investigation → independent attempt.

MAIN LESSON · SECTION 1

Draw the comparison with honest scales and visible denominators

> Preserve units, case counts and uncertainty limits without making a figure stand in for missing evidence.

Learning goals

I can explain the **mechanism** and terminology of figures that preserve **evidence**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Negative results and claim revision. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Inspecting failures and anomalies. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Power and resource tradeoffs. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Starting explanation

- Recall Negative results and claim revision. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

End goal

Explain figures that preserve **evidence** using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of figures that preserve **evidence**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Understand the question and evidence contract

> Preserve units, case counts and uncertainty limits without making a figure stand in for missing evidence.

Start with the research problem

A chart changes how easy a difference is to see; it does not add observations. Label what the axis measures, how many cases contributed and which conditions were evaluated. For ordinary magnitude bars, a zero baseline preserves the meaning of bar length. A focused difference plot can use another clearly explained scale, but hidden truncation makes a small difference look much larger.

Keep missing uncertainty missing rather than decorating a plot with invented error bars.

Original concept explanation

Figures should preserve denominators, scales and uncertainty rather than exaggerate the preferred comparison. A plot is a representation of **evidence**, not extra **evidence**.

Complete case inputs, rules and limits

Figure item	Complete supplied exercise contract
Main rates	A 80%, B 82%; raw case counts are not supplied
Main problematic axis	Bars start at 79% without adequate disclosure
Correct main difference	Two percentage points; not a several-fold increase in the actual rates
Separate picture	A six correct, B eight correct on the same ten cases
Picture axis	Correct-count axis starts at zero; denominator ten visible
Uncertainty	No interval or raw study design supplied; do not fabricate error bars
Interpretation	A clear figure does not establish sampling quality, causal advantage or population performance

Name given and missing evidence

List the given facts and one fact that is not supplied.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Trace the complete main case before interpreting it

> Follow each evidence-to-conclusion step and retain unknown quantities.

Original declared procedure

Label axes, units, cases and scope. For simple comparative bars use a zero baseline or explicitly justify another plot. Show condition breakdowns and unassessed counts where relevant.

Complete main-case evidence ledger

Main figure calculation	Complete value	Meaning
A rate	80%	Supplied rate; underlying denominator unknown
B rate	82%	Supplied rate; underlying denominator unknown
Absolute difference	82% – 80% = 2 percentage points	Correct rate-scale difference
Relative increase	2/80 = 2.5%	Different from two percentage points
Heights above hidden 79% baseline	A: 1 point; B: 3 points	Threefold drawn heights distort the actual rate ratio
Repair	Zero-baseline percentage bars or clearly labelled difference plot	Retain missing case-count and uncertainty information

State the scope and explain the visual

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Figures that preserve evidence — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Rates A 80%, B 82% are plotted on bars starting 79%, making B look several times larger.	Label axes, units, cases and scope. For simple comparative bars use a zero baseline or explicitly justify another plot. Show condition breakdowns and unassessed counts where relevant.	Difference is 2 percentage points; truncated bars exaggerate magnitude. Use zero-baseline bars or a clearly labelled difference plot.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

Predict the changed case before reading the explanation

> Keep the rule and evidence boundary visible.

Main task and response

- Rates A **80%**, B 82% are plotted on bars starting 79%, making B look several times larger. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured change

- Change: Start the axis at 5 without making the truncation obvious.

▼ Check after predicting

Prediction: The bars exaggerate the visual contrast and require correction or clear disclosure.

Reason: Visible heights no longer correspond to the full count scale.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

Connect the main result with the separate pictured case

> Keep the two scenarios and their supplied evidence distinct.

Main case and original checked reasoning

Rates A **80%**, B 82% are plotted on bars starting 79%, making B look several times larger.

Difference is 2 percentage points; truncated bars exaggerate magnitude. Use zero-baseline bars or a clearly labelled **difference plot**.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate pictured inputs and complete explanation

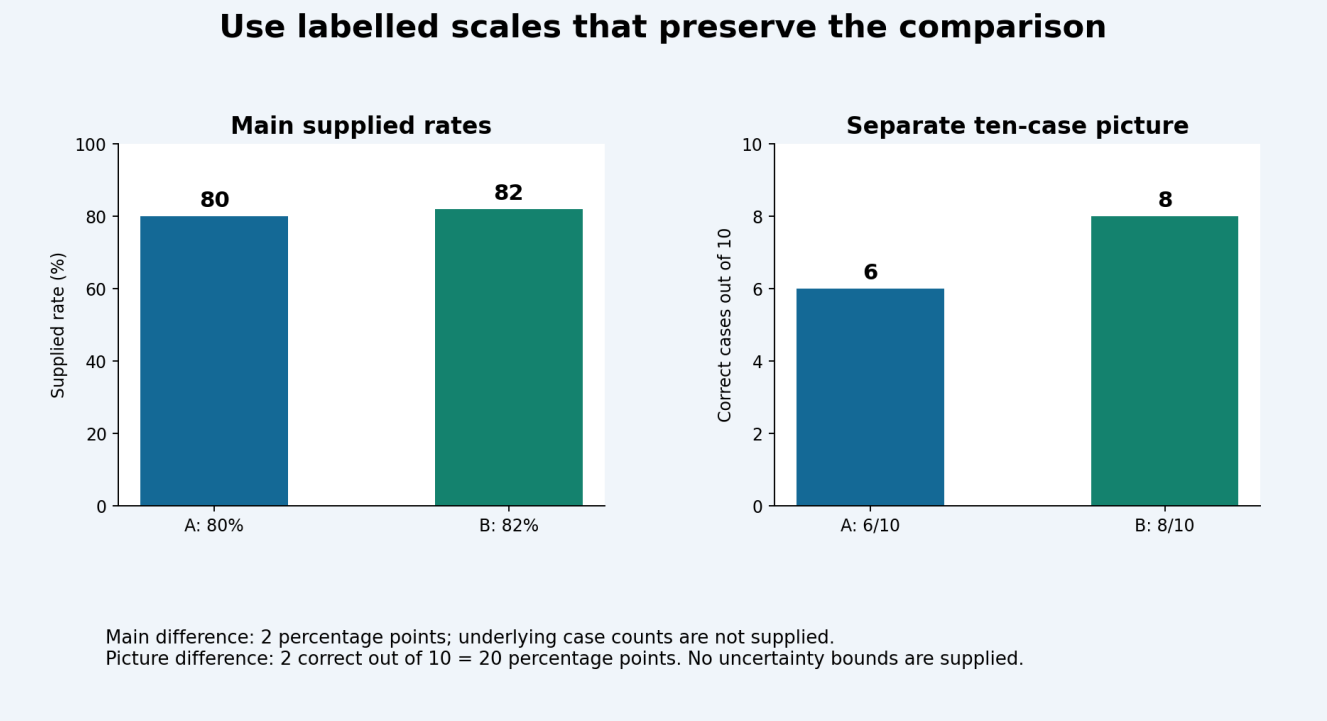
Draw bars that preserve the comparison. A report plots correct counts for two versions on the same ten cases.

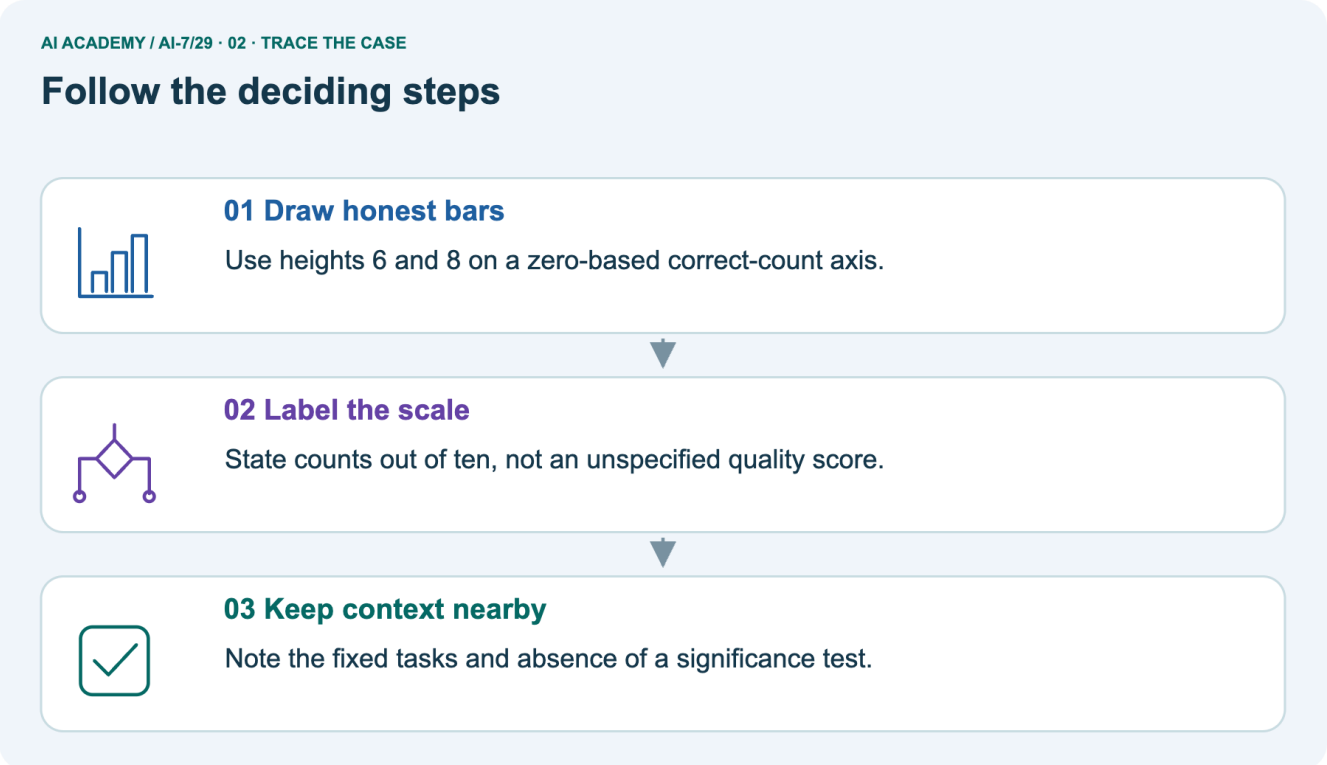
A has 6/10 correct; B has 8/10. The count axis should start at zero and identify the ten-case denominator.

The separate picture compares six versus eight correct on the same ten cases. A is 60%, B 80%; the difference is two correct or twenty percentage points. Use count bars from zero to ten and label the common ten-case denominator. If the bars secretly start at five, their visible heights are one and three, creating a threefold height ratio even though the actual count ratio is eight divided by six.

The new native figure keeps main supplied percentages separate from the ten-case count example. No uncertainty bounds are supplied, and a well-labelled chart does not turn ten cases into a population or causal conclusion.

Read the labelled representation





Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Draw honest bars	Use heights 6 and 8 on a zero-based correct-count axis.
2. Label the scale	State counts out of ten, not an unspecified quality score.
3. Keep context nearby	Note the fixed tasks and absence of a significance test.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Figures make **evidence** easier to inspect when labels and scales preserve meaning. A dramatic-looking visual cannot supply missing uncertainty or representativeness.

Complete picture evidence and limits

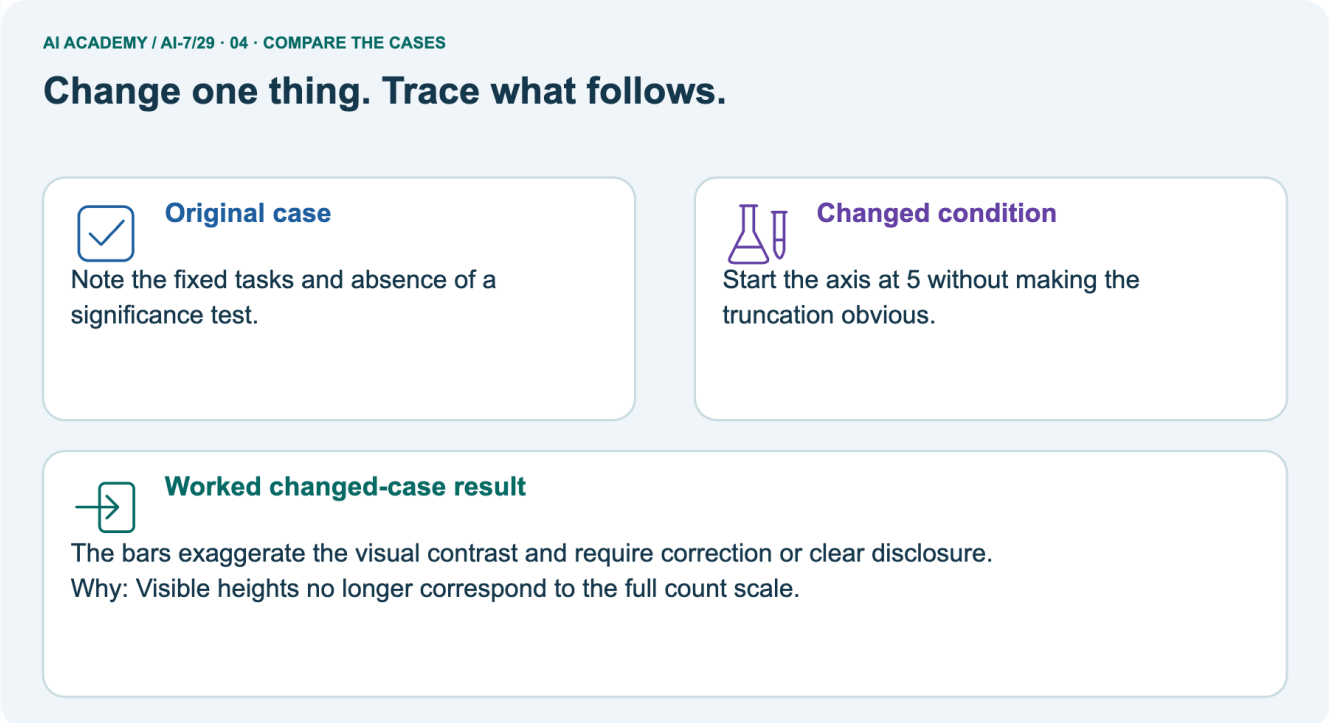
Separate ten-case picture	A	B
Correct count	6	8
Evaluated cases	10	10
Observed rate	60%	80%
Difference	Two extra correct	Twenty percentage points
Zero-baseline count height	6	8

Separate ten-case picture	A	B
Hidden start-at-five height	1	3
Why hidden truncation misleads	Drawn heights appear threefold	Actual count ratio 8/6 is not threefold
Uncertainty / population inference	Not supplied	Not supplied

Predict the changed condition

- Change one thing: Start the axis at 5 without making the truncation obvious. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain what remains unknown

- A well-labelled chart still cannot establish a causal or population claim from ten cases.

- Read the labelled picture: Draw count bars for 6/10 and 8/10 using the stated axis. Why must the denominator and fixed task context remain visible, and what distortion arises from an undisclosed axis start at five?

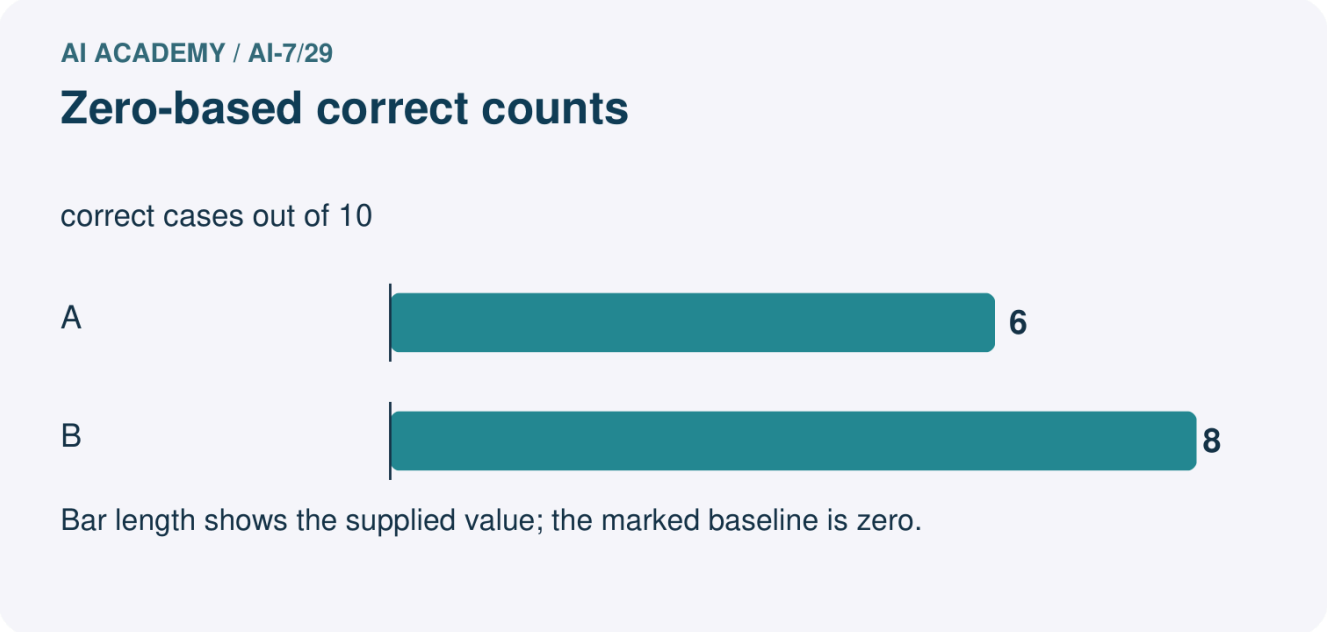
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Figures that preserve evidence

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/29

Figures that preserve evidence

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Draw honest bars

Use heights 6 and 8 on a zero-based correct-count axis.





2. Label the scale

State counts out of ten, not an unspecified quality score.





3. Keep context nearby

Note the fixed tasks and absence of a significance test.

CHANGE ONE THING

Start the axis at 5 without making the truncation obvious.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Find the first unsupported inference and repair it

> Use the supplied conditions rather than inventing missing evidence.

Original qualification and repair setup

Visual choices must reflect the intended metric and cannot imply causal certainty from association.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete boundary and misconception probes

Figure audit	Checked interpretation
Main 80% versus 82%	Two percentage points, not two percent relative
Underlying main denominator missing	Do not invent eighty or eighty-two of one hundred cases
Picture counts six/eight with common ten	Show denominator and zero count baseline
Bar axis truncated without clear disclosure	Exaggerates relative bar lengths
Focused difference plot	Use an explicit difference scale and explain units
Uncertainty not supplied	Do not invent intervals or significance stars
Counts eight versus nine with different totals	Show rates and both denominators
Different case sets	Rate comparison alone does not prove a fair controlled advantage

Write and justify the repair

- A report omits this qualification: Visual choices must reflect the intended metric and cannot imply causal certainty from association.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise the mechanism with progressively less help

> Keep the selected independent case for a first attempt.

Shared explanation

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Supported workbook trace

Use workbook W2 for the connected example “Figures that preserve **evidence**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual scope check

- Use the worked case for Figures that preserve **evidence**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Draw count bars for 6/10 and 8/10 using the stated axis. Why must the denominator and fixed task context remain visible, and what distortion arises from an undisclosed axis start at five?

> **My explanation using a pictured step and the given facts**

Reserve the independent task

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

Investigate, record the evidence and check the limits

> Distinguish a prediction, a paper trace and an actual local execution.

Current investigation

Rates A **80%**, B 82% are plotted on bars starting 79%, making B look several times larger. Show the intermediate steps and give the conclusion with its scope.

Complete investigation instructions

1. Perform a paper figure audit: write main rates, their two-percentage-point difference and missing denominator. Draw repaired zero-baseline percentage bars, or an explicitly labelled difference plot. Then draw the separate six/eight-of-ten count chart from zero to ten. Calculate the misleading heights above a hidden baseline of five and explain why drawn-height ratio is not actual count ratio.

   Keep missing uncertainty and study scope visible. No real plotting data, interval or causal advantage is invented. Reserve the unequal-denominator independent case for the first attempt.

Evidence record and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try the selected new case independently

> Correct the unequal-denominator plot using both rates and totals, then state what comparison conditions remain unknown.

Complete independent task

Counts A 8/10, B 9/15 are plotted as 8 versus 9 without denominators. Correct.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Attempt this without hints

Counts A 8/10, B 9/15 are plotted as 8 versus 9 without denominators. Correct.

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save the research artifact and unresolved conditions

> Keep supplied evidence, proposed revisions and actual work distinct.

Project task and artifact

Add a labelled figures that preserve **evidence evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of figures that preserve **evidence**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence and limitations

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Retrieve the mechanism and state the evidence boundary

> Explain what follows and what would require new evidence.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of figures that preserve **evidence**

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use research vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use research vocabulary precisely

> Connect each term to the question, comparison, measurement or evidence.

Vocabulary and your explanation

Word	Meaning
Axis scale	The mapping between values and plotted positions
Denominator label	The sample size underlying a proportion
Difference plot	A display of a defined comparison magnitude

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: Visual choices must reflect the intended metric and cannot imply causal certainty from association.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the research reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the research reasoning and its limits

> Use after practice; reserve the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Figures that preserve evidence — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Rates A 80%, B 82% are plotted on bars starting 79%, making B look several times larger.	Label axes, units, cases and scope. For simple comparative bars use a zero baseline or explicitly justify another plot. Show condition breakdowns and unassessed counts where relevant.	Difference is 2 percentage points; truncated bars exaggerate magnitude. Use zero-baseline bars or a clearly labelled difference plot.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Draw bars that preserve the comparison. A report plots correct counts for two versions on the same ten cases.

By the end, I can	How I will show it
Explain the mechanism and terminology of figures that preserve evidence	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Draw bars that preserve the comparison. A report plots correct counts for two versions on the same ten cases.
- Which labelled step matches this detail? Note the fixed tasks and absence of a significance test.
- 1. Draw honest bars
- 2. Label the scale
- 3. Keep context nearby
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Keep context nearby Note the fixed tasks and absence of a significance test. Figures make **evidence** easier to inspect when labels and scales preserve meaning.

A dramatic-looking visual cannot supply missing uncertainty or representativeness.

Changed-case question and optional hint

- Changed-case practice: Start the axis at 5 without making the truncation obvious. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. The bars exaggerate the visual contrast and require correction or clear disclosure.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The bars exaggerate the visual contrast and require correction or clear disclosure. Visible heights no longer correspond to the full count scale.

Repair and retain the record

Review Negative results and claim revision (ai-7/28). Goal: Explain the **mechanism** and terminology of negative results and claim revision. Short practice: Does a 6/8 tie support the **prediction** that B scores higher?

Explain why one newly selected favourable demonstration does not replace the original result or prove equivalence.. Return to this week and explain what you changed.

Review Inspecting failures and anomalies (ai-7/26). Goal: Explain the **mechanism** and terminology of inspecting failures and anomalies. Short practice: Calculate mean and median for 2,2,2,10.

What does the timeout log establish, and why is deleting that inconvenient record incompatible with the stated plan?. Return to this week and explain what you changed.

Review Power and resource tradeoffs (ai-7/23). Goal: Explain the **mechanism** and terminology of power and resource tradeoffs. Short practice: Calculate both simulated detection fractions and resource costs.

What additional detection is observed for the extra forty units, and does that establish power for an actual classroom study?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

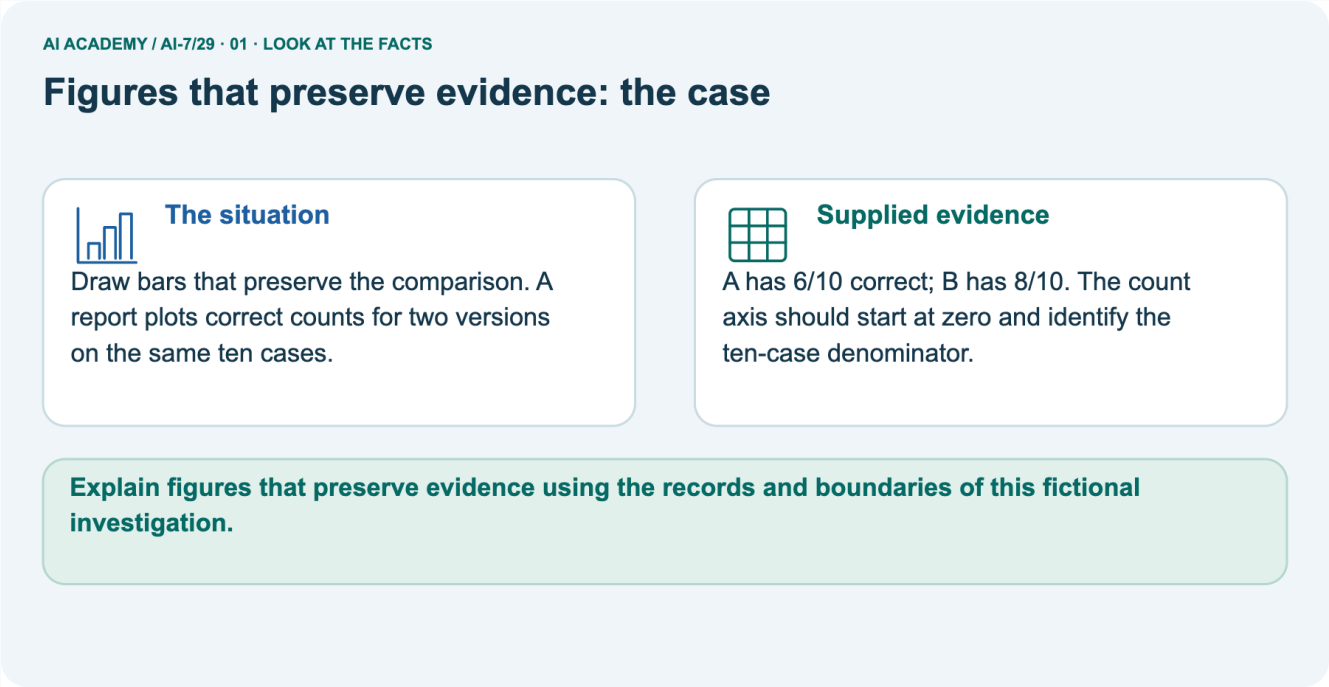
▼ **Reference · Six-picture concept journey**

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Figures that preserve evidence: the case



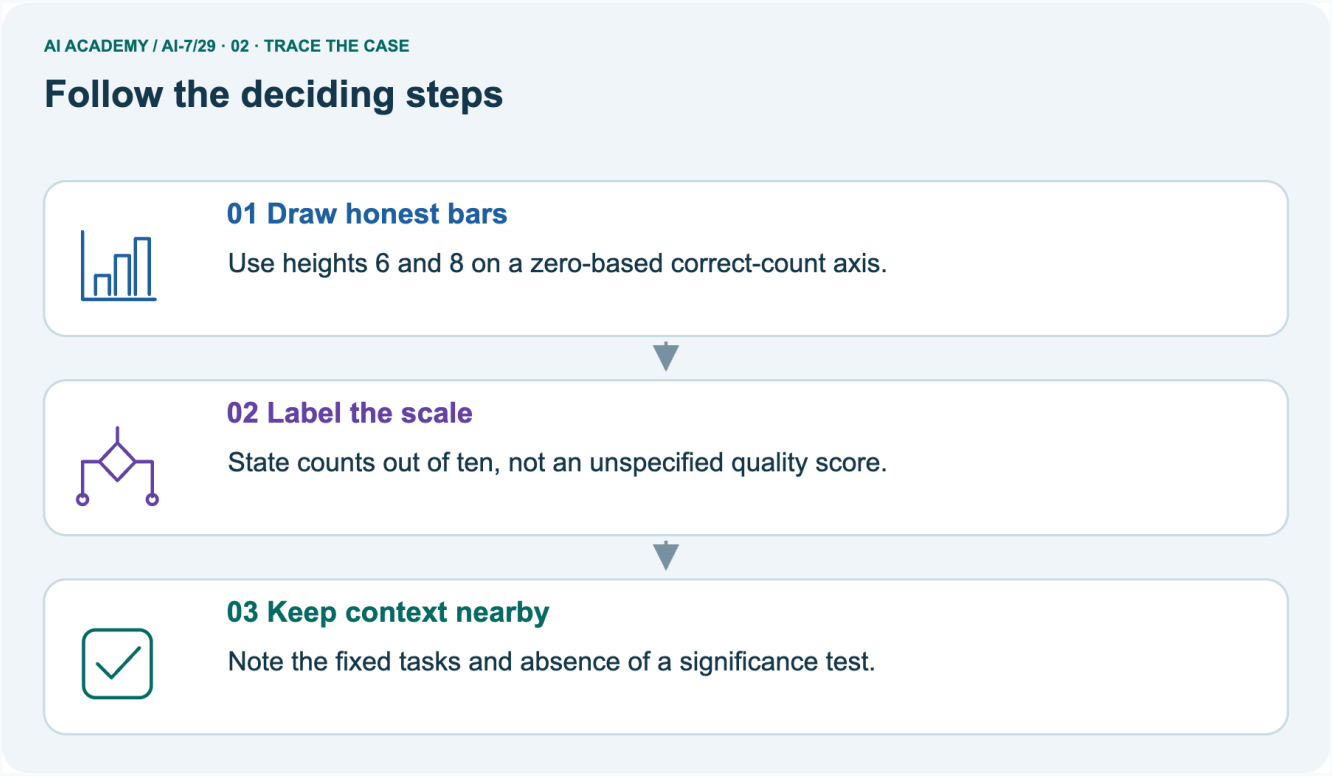
Follow the labelled picture with the matching explanation. Open the vector diagram

Draw bars that preserve the comparison. A report plots correct counts for two versions on the same ten cases. Supplied facts: A has 6/10 correct; B has 8/10.

The count axis should start at zero and identify the ten-case denominator.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

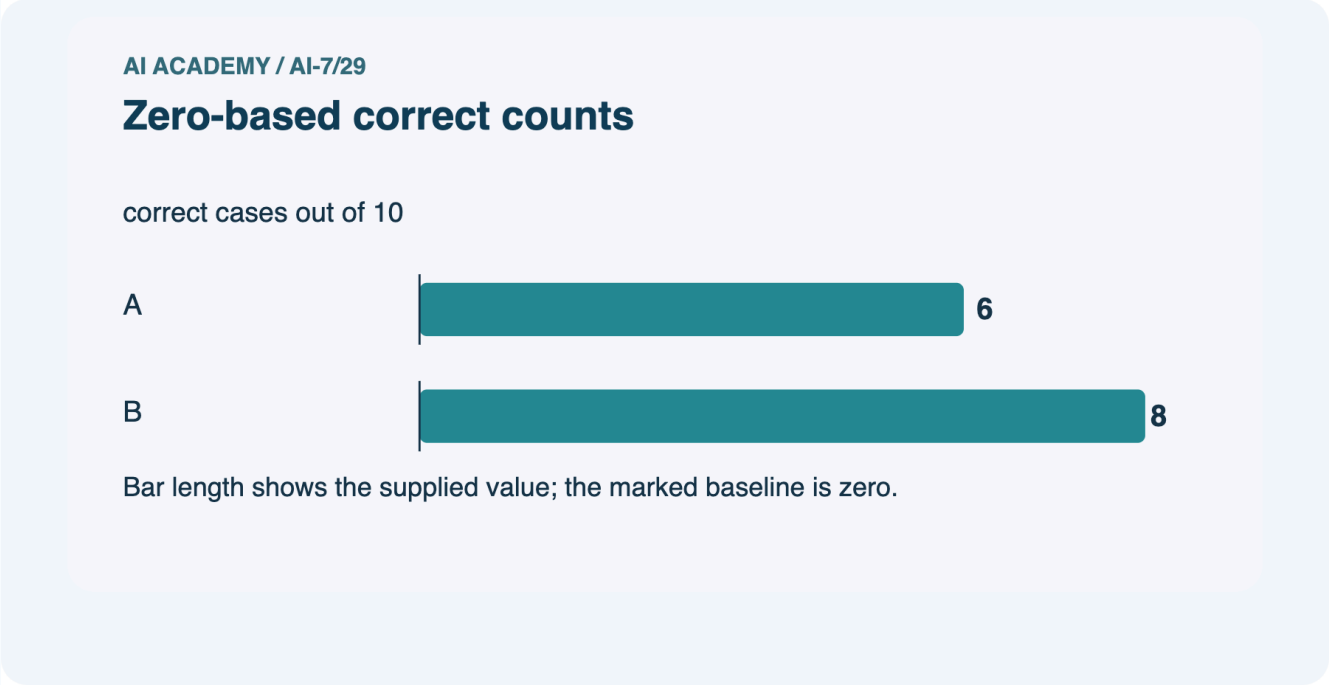


Follow the labelled picture with the matching explanation. Open the vector diagram

Draw honest bars: Use heights 6 and 8 on a zero-based correct-count axis. → Label the scale: State counts out of ten, not an unspecified quality score. → Keep context nearby: Note the fixed tasks and absence of a significance test.

Point to each numbered stage and explain which recorded **input** it uses.

03. Zero-based correct counts

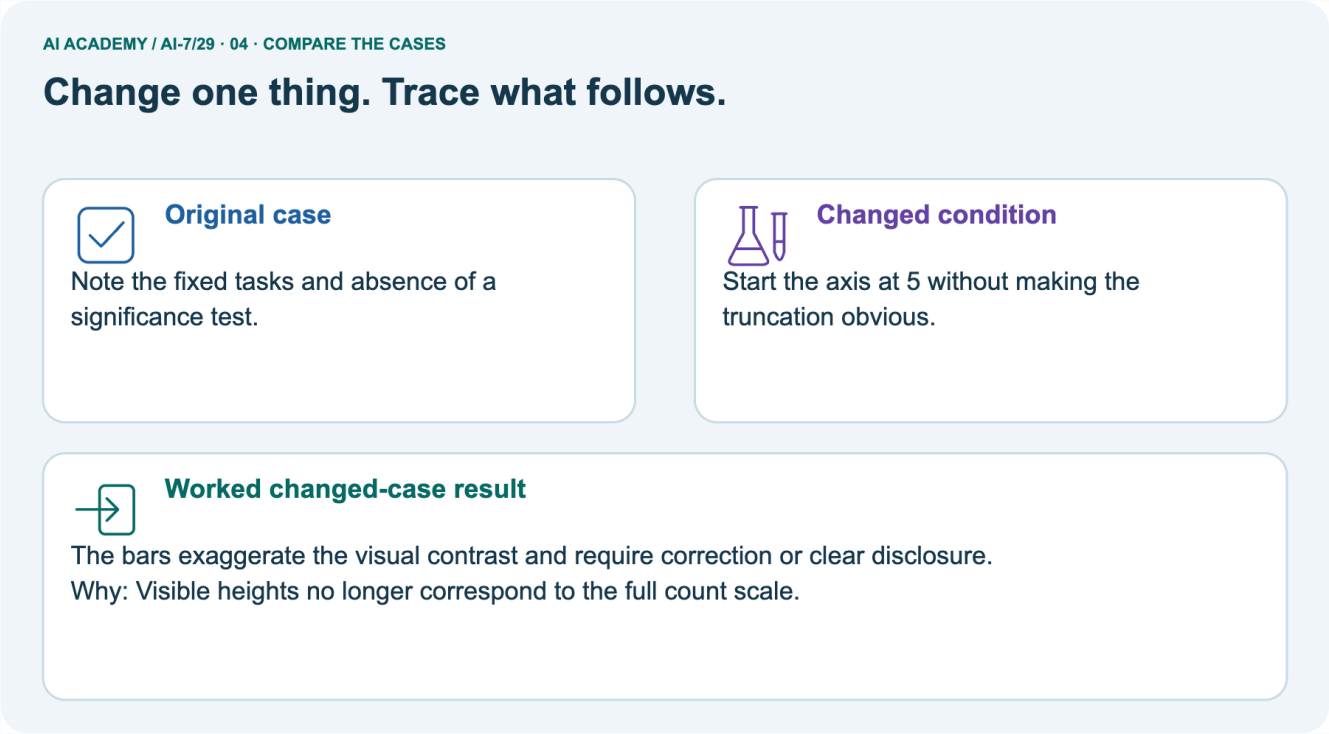


Follow the labelled picture with the matching explanation. Open the vector diagram

Zero-based correct counts. Units: correct cases out of 10. A: 6. B: 8.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



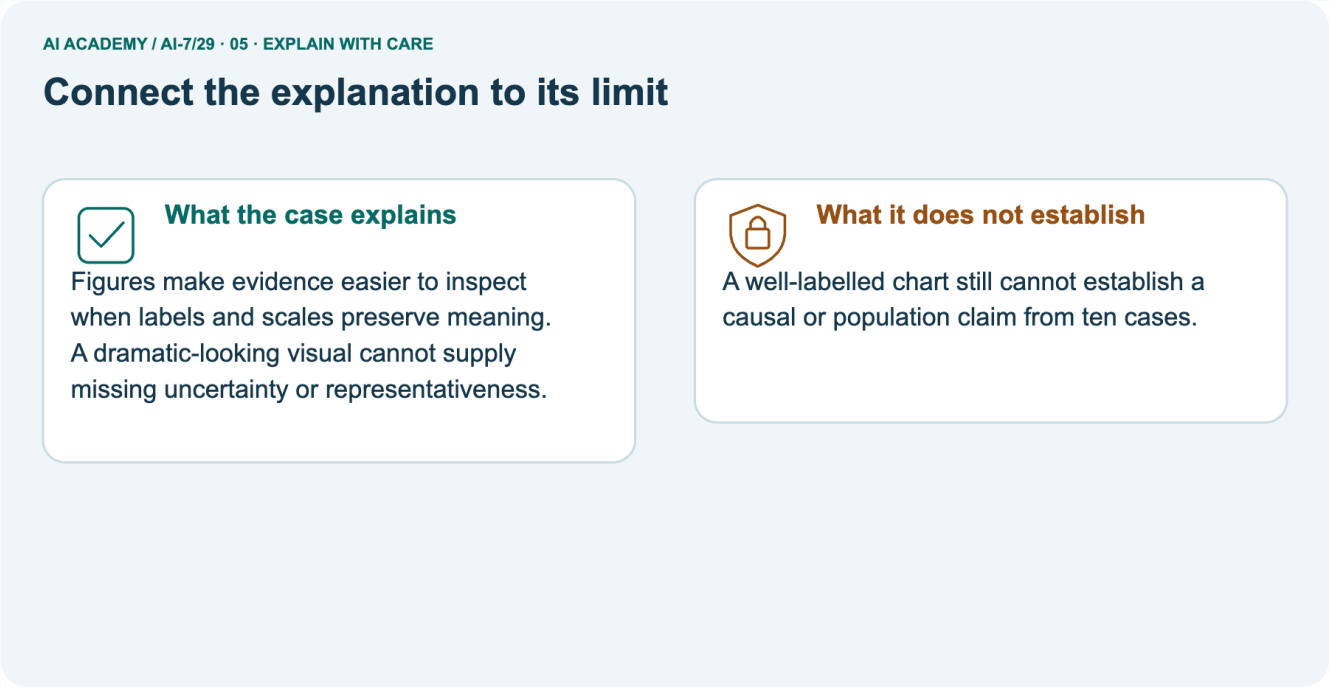
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Note the fixed tasks and absence of a significance test. Changed condition: Start the axis at 5 without making the truncation obvious.

Expected result: The bars exaggerate the visual contrast and require correction or clear disclosure. Reason: Visible heights no longer correspond to the full count scale.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



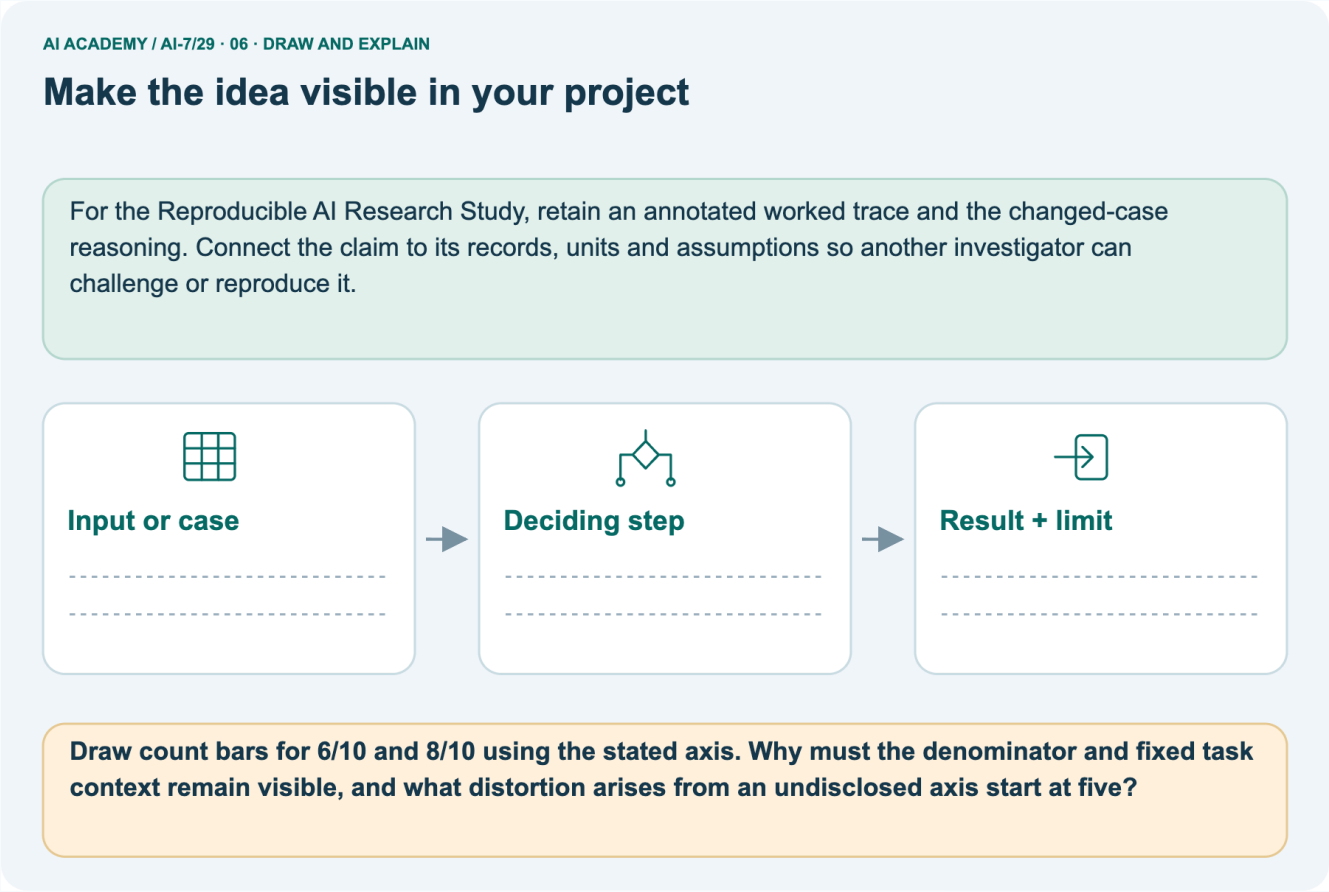
Follow the labelled picture with the matching explanation. Open the vector diagram

Figures make **evidence** easier to inspect when labels and scales preserve meaning. A dramatic-looking visual cannot supply missing uncertainty or representativeness.

Limit: A well-labelled chart still cannot establish a causal or population claim from ten cases.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Draw count bars for 6/10 and 8/10 using the stated axis.

Why must the denominator and fixed task context remain visible, and what distortion arises from an undisclosed axis start at five?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Figures that preserve evidence: the case

02. Follow the deciding steps

03. Zero-based correct counts

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.